

EXPLANATIONS

1. (c) Resultant vector,

$$\vec{R} = \sqrt{a^2 + b^2 + 2ab\cos\theta}$$

According to question, $a = b = R$

$$\vec{R} = \sqrt{R^2 + R^2 + 2R^2\cos\theta}$$

$$= R\sqrt{2+2\cos\theta}$$

$$= \sqrt{2}R\sqrt{2\cos^2\frac{\theta}{2}} = 2R\cos\frac{\theta}{2}$$

2. [4] $\vec{A} = 3\hat{i} + 4\hat{j}$ & $\vec{B} = 4\hat{i} + 3\hat{j}$

$$\vec{N} = |\vec{A}| \cdot \vec{B} = \frac{5(4\hat{i} + 3\hat{j})}{5} = 4\hat{i} + 3\hat{j}$$

$$\therefore x = 4$$

3. [6] $|\vec{F}_R| = |\vec{F}_2| = 3F$

$$|\vec{F}_1| = F$$

$$\text{Now, } F_R^2 = F_1^2 + F_2^2 + 2F_1F_2\cos\theta$$

$$\Rightarrow 9F^2 = F^2 + 9F^2 + 6F^2\cos\theta$$

$$\Rightarrow \cos\theta = -\frac{1}{6}$$

$$\Rightarrow \theta = \cos^{-1}\left(-\frac{1}{6}\right)$$

$$\therefore n = -6 \text{ and } |n| = 6$$

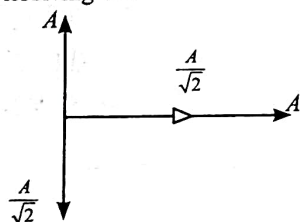
4. (a) Let the resultant of both the vectors be $\vec{R}$, then

$$\vec{R} = (2\vec{Q} + 2\vec{P}) + (2\vec{Q} - 2\vec{P})$$

$$\vec{R} = 4\vec{Q}$$

Angle between $\vec{Q}$ and $\vec{R}$ is zero

5. [3] Resolving $\vec{OR}$



the resultant of the three vectors,

$$\vec{R} = \left(A + \frac{A}{\sqrt{2}}\right)\hat{i} + \left(A - \frac{A}{\sqrt{2}}\right)\hat{j}$$

the resultant of the three vectors,

$$|\vec{R}| = \sqrt{\left(A + \frac{A}{\sqrt{2}}\right)^2 + \left(A - \frac{A}{\sqrt{2}}\right)^2} = \sqrt{3}A$$

6. [3] Given: $\cos\theta = \frac{5}{9}$

$$\Rightarrow \frac{\vec{a} \cdot \vec{b}}{ab} = \frac{5}{9}$$

...(i)

$$\Rightarrow |\vec{a} + \vec{b}| = \sqrt{2}|\vec{a} - \vec{b}|$$

$$\Rightarrow a^2 + b^2 + 2\vec{a} \cdot \vec{b} = 2a^2 + 2b^2 - 4\vec{a} \cdot \vec{b}$$

$$\Rightarrow 6\vec{a} \cdot \vec{b} = a^2 + b^2$$

$$\Rightarrow 6 \times \frac{5}{9}ab = a^2 + b^2$$

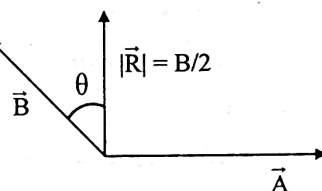
$$\Rightarrow \frac{10}{3}ab = a^2 + b^2 \text{ \& } a = nb$$

$$\Rightarrow 3n^2 - 10n + 3 = 0$$

$$\Rightarrow n = \frac{1}{3} \text{ and } n = 3$$

Thus, $n = 3$

7. [150]



According to the question.

$$B\cos\theta = \frac{B}{2} \Rightarrow \theta = 60^\circ$$

Hence, the angle between $\vec{A}$ & $\vec{B}$ is $90^\circ + 60^\circ = 150^\circ$

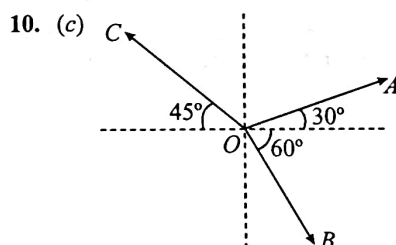
8. (*) Given,

$$\vec{B} - \vec{A} = 2\hat{j}, \vec{B} = 2\hat{j} + 2\hat{i} + 3\hat{j} + 2\hat{k}$$

$$\therefore \vec{B} = 2\hat{i} + 5\hat{j} + 2\hat{k}$$

$$|\vec{B}| = \sqrt{(2)^2 + (5)^2 + (2)^2} = \sqrt{33} \text{ (Bonus)}$$

9. (b) Apply the triangle law of addition, to get answer.



According to diagram, let magnitude be equal to

$$\vec{OA} = \lambda \left[\cos 30^\circ \hat{i} + \sin 30^\circ \hat{j} \right] = \lambda \left[\frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j} \right]$$

$$\vec{OB} = \lambda \left[\cos 60^\circ \hat{i} - \sin 60^\circ \hat{j} \right] = \lambda \left[\frac{1}{2} \hat{i} - \frac{\sqrt{3}}{2} \hat{j} \right]$$

$$\vec{OC} = \lambda \left[\cos 45^\circ (-\hat{i}) + \sin 45^\circ \hat{j} \right] = \lambda \left[-\frac{1}{\sqrt{2}} \hat{i} + \frac{1}{\sqrt{2}} \hat{j} \right]$$

$$\therefore \vec{OA} + \vec{OB} - \vec{OC}$$

$$= \lambda \left[\left(\frac{\sqrt{3}+1}{2} + \frac{1}{\sqrt{2}} \right) \hat{i} + \left(\frac{1}{2} - \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \right) \hat{j} \right]$$

$$\tan^{-1} \left[\frac{\frac{1}{2} - \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}}{\frac{\sqrt{3}+1}{2} + \frac{1}{\sqrt{2}}} \right] = \tan^{-1} \left[\frac{\sqrt{2}-\sqrt{6}-2}{\sqrt{6}+\sqrt{2}+2} \right]$$

$$= \tan^{-1} \left[\frac{1-\sqrt{3}-\sqrt{2}}{\sqrt{3}+1+\sqrt{2}} \right]$$

11. (a) $\vec{AB} + \vec{AH} = \vec{AO} \Rightarrow \vec{AC} + \vec{AG} = 2\vec{AO}$

$$\vec{AD} + \vec{AF} = 3\vec{AO} \Rightarrow \vec{AE} = 2\vec{AO}$$

Adding all,

$$\vec{AB} + \vec{AC} + \vec{AD} + \vec{AE} + \vec{AF} + \vec{AG} + \vec{AH} = 8\vec{AO}$$

$$= 16\hat{i} + 24\hat{j} - 32\hat{k}$$

$$\vec{AB} + \vec{AO} + \vec{OB}, \vec{AC}$$

$$= \vec{AO} + \vec{OC}, \vec{AD}, \vec{AO} + \vec{OD}$$

$$\vec{AE} = 2\vec{AO}, \vec{AF}$$

$$= \vec{AO} + \vec{OF}, \vec{AG} = \vec{AO} + \vec{OG}$$

$$\vec{AH} = \vec{AO} + \vec{OH}$$

Adding all we get $8\vec{AO}$

12. (a) The horizontal component of force,

$$\vec{F}_x = \left[10 \times \frac{\sqrt{3}}{2} + 20 \times \frac{1}{2} + \frac{20}{\sqrt{2}} - \frac{15}{\sqrt{2}} - \frac{15\sqrt{3}}{2} \right]$$

$$= 9.25\hat{i}$$

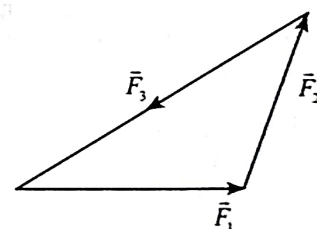
The vertical component of force,

$$\vec{F}_y = \left[15 \times \frac{1}{2} + 20 \times \frac{\sqrt{3}}{2} + 10 \right]$$

$$\times \frac{1}{2} - \frac{15}{\sqrt{2}} - \frac{20}{\sqrt{2}} = 5\hat{j}$$

$$\therefore \vec{F}_R = \vec{F}_x + \vec{F}_y = 9.25\hat{i} + 5\hat{j}$$

13. (b)



$\therefore$ For equilibrium $F_{\text{net}} = 0$

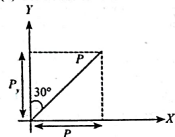
14. (d) $\vec{F} = (\vec{F}_1 + \vec{F}_2) \Rightarrow |\vec{F}_1| = A, |\vec{F}_2| = \frac{A}{2}$

$$\theta = 90^\circ$$

$$\therefore |\vec{F}| = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$$

$$|\vec{F}| = \sqrt{A^2 + \frac{A^2}{4}} = \frac{A\sqrt{5}}{2}$$

15. (d) Let the vector be P .



$$P_x = P \cos 30^\circ = 2\sqrt{3}$$

$$\Rightarrow P \frac{\sqrt{3}}{2} = 2\sqrt{3} \Rightarrow P = 4$$

$$\text{Now } P_y = P \sin 30^\circ = 4 \times \frac{1}{2} = 2$$

16. (a) $\vec{r} = (0.5\hat{i} + 3\hat{j} + 9\hat{k}) \text{ m/s}$

$$\text{At } t = 2\text{s}$$

$$\vec{r} = (2\hat{i} + 6\hat{j} + 9\hat{k})$$

Direction cosine along y-axis,

$$\cos \theta = \frac{6}{\sqrt{2^2 + 6^2 + 9^2}} = \frac{6}{\sqrt{121}} = \frac{6}{11}$$

$$\therefore \sin \theta = \frac{\sqrt{85}}{11}$$

$$\text{and } \tan \theta = \frac{\sqrt{85}}{6}$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{85}}{6}\right)$$

from y-axis.

17. (d) $\vec{AB} = \vec{AO} + \vec{OB}, \vec{AC}$

$$= \vec{AO} + \vec{OC}, \vec{AD}$$

$$= \vec{AO} + \vec{OD} = 2\vec{AO}$$

$$\Rightarrow \vec{AB} + \vec{AC} + \vec{AD} = 4\vec{AO} + \vec{OB} + \vec{OC}$$

18. (c) $G = \left(\frac{a}{2}, 0, \frac{a}{2}\right), H = \left(0, \frac{a}{2}, \frac{a}{2}\right)$

$$\vec{GH} = -\frac{a}{2}\hat{i} + \frac{a}{2}\hat{j} = \frac{a}{2}(\hat{j} - \hat{i})$$

19. [3] When two vectors are perpendicular, their dot product is zero. That is:

$$\vec{A} \cdot \vec{B} = 0$$

$$4 - 6n + 8p = 0$$

Now, the magnitudes of both vectors should be equal:

$$|\vec{A}| = |\vec{B}|$$

$$4 + 9n^2 + 4 = 4 + 4 + 16p^2$$

$$9n^2 = 16p^2$$

$$P = \frac{3}{4}n$$

$$4 - 6n + 6n = 0, 12n = 4$$

$$n = \frac{1}{3}$$

20. (d) As $\vec{P}$ and $\vec{Q}$ are perpendicular their dot product must be zero.

$$\vec{P} \cdot \vec{Q} = 0$$

$$(i + 2mj + mk) \cdot (4i - 2j + mk) = 0$$

$$\Rightarrow 4 - 4m + m^2 = 0 \Rightarrow m = 2$$

21. [1] For two perpendicular vectors

$$(ai + bj + k) \cdot (2i - 3j + 4k) = 0$$

$$2a - 3b + 4 = 0 \quad \dots(i)$$

$$\text{On solving, } 2a - 3b = -4$$

$$\text{Also given } 3a + 2b = 7 \quad \dots(ii)$$

Form (i) & (ii) We get $a = 1, b = 2$

$$\frac{1}{2} = \frac{a}{b} \times \frac{x}{2} \Rightarrow x = 1$$

22. (b)

$$|\vec{A} + \vec{B}| = \sqrt{|\vec{A}|^2 + |\vec{B}|^2 + 2|\vec{A}||\vec{B}|\cos\theta}$$

$$= \sqrt{1+1+2\cos\theta} = 2\cos\theta/2$$

$$|\vec{A} - \vec{B}| = \sqrt{|\vec{A}|^2 + |\vec{B}|^2 - 2|\vec{A}||\vec{B}|\cos\theta}$$

$$= \sqrt{2-2\cos\theta} = 2\sin\theta/2$$

$$\frac{|\vec{A} + \vec{B}|}{|\vec{A} - \vec{B}|} = \cot\theta/2$$

$$\Rightarrow |\vec{A} - \vec{B}| = |\vec{A} + \vec{B}| \tan \frac{\theta}{2}$$

23. (c) According to the question

$$|\vec{A} + \vec{B}| = 2|\vec{A} - \vec{B}| \quad \dots(i)$$

Squaring equation (i) both side, $[\because A = B]$

$$|\vec{A} + \vec{B}|^2 = (2|\vec{A} - \vec{B}|)^2$$

$$\Rightarrow 2A^2 + 2A^2 \cos\theta = 4(A^2 - 2A^2 \cos\theta)$$

$$\Rightarrow \cos\theta = \frac{3}{5}$$

$$\therefore \theta = \cos^{-1}(3/5)$$

24. [2] The magnitude of the component of vector $\vec{A}$ along vector $\vec{B}$ will be

$$= \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} = \frac{2+6-2}{3} = 2$$

25. [5] $0 = (2\hat{i} + 4\hat{j} - 2\hat{k}) \cdot \left(\frac{\hat{i} + 2\hat{j} + a\hat{k}}{\sqrt{1+4+a^2}}\right)$

$$0 = 2 + 8 - 2a \Rightarrow a = 5$$

26. (a) Given that, $\vec{A} \cdot \vec{B} = |\vec{A} \times \vec{B}|$

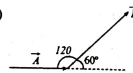
$$\Rightarrow AB \cos\theta = AB \sin\theta \Rightarrow \theta = 45^\circ$$

$$|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB \cos\theta}$$

$$= \sqrt{A^2 + B^2 - 2AB \times \frac{1}{\sqrt{2}}}$$

$$= \sqrt{A^2 + B^2 - \sqrt{2}AB}$$

27. (a)



According to diagram,

Angle between $\vec{A}$ and $\vec{B}$, $\theta = 60^\circ$

Angle between $\vec{A}$ and $-\vec{B}$, $\theta = 120^\circ$

If angle between $\vec{A}$ and $\vec{A} - \vec{B}$ is α

$$\text{then } \tan \alpha = \frac{|\vec{B}| \sin \theta}{|\vec{A}| + |\vec{B}| \cos \theta}$$

$$= \frac{B \sin 120^\circ}{A + B \cos 120^\circ} = \frac{B \frac{\sqrt{3}}{2}}{A - \frac{B}{2}}$$

$$\Rightarrow \tan \alpha = \frac{\sqrt{3}B}{2A - B}$$

Hence, the angle between vector $\vec{A}$ and

$$(\vec{A} - \vec{B}) \text{ is } \tan \alpha = \frac{\sqrt{3}B}{2A - B}$$

28. (c) Projection of $\vec{A}$ on $\vec{B} = (\vec{A} \cdot \vec{B})/\vec{B}$

$$= \left[\left(\frac{i+j}{\sqrt{2}} \right) \cdot \left(\frac{i+j}{\sqrt{2}} \right) \right] \frac{(i+j)}{\sqrt{2}}$$

$$= \frac{1}{2} (1+1) (i+j)$$

$$= \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) \frac{(i+j)}{\sqrt{2}} = i+j$$

29. (a) $|\vec{A} + \vec{B}| = n|\vec{A} - \vec{B}|$

$$\Rightarrow A^2 + B^2 + 2AB \cos \theta$$

$$= n^2(A^2 + B^2 - 2AB \cos \theta)$$

$$\Rightarrow \cos \theta (1 + n^2) = \frac{2A^2(n^2 - 1)}{2A^2}$$

$$[A = B = a]$$

$$\cos \theta = \frac{n^2 - 1}{n^2 + 1}$$

30. (c) Given, $|\vec{x}| = |\vec{y}|$ and $|\vec{x} - \vec{y}| = n|\vec{x} + \vec{y}|$

$$x^2 + y^2 - 2\vec{x} \cdot \vec{y} = n^2(x^2 + y^2 + 2\vec{x} \cdot \vec{y})$$

$$(1 - n^2)(x^2 + y^2) = (1 + n^2)2\vec{x} \cdot \vec{y}$$

$$(1 - n^2)(x^2 + y^2) = (1 + n^2)2xy \cos \theta$$

$$\cos \theta = \frac{1 - n^2}{1 + n^2} \therefore 2xy = x^2 + y^2$$

$$\theta = \cos^{-1}\left(\frac{n^2 - 1}{n^2 + 1}\right)$$

31. (a) According to question the given data

is two forces $(\vec{P} + \vec{Q})$ and $(\vec{P} - \vec{Q})$

$$\vec{A} = \vec{P} + \vec{Q} \text{ and } \vec{B} = \vec{P} - \vec{Q} \text{ and } \vec{P} \perp \vec{Q}$$

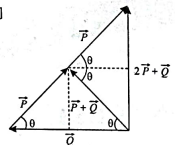
$$\Rightarrow |\vec{A}| = |\vec{B}| = \sqrt{2(P^2 + Q^2)} (1 + \cos \theta)$$

$$\Rightarrow \text{For } |\vec{A}| = |\vec{B}| = \sqrt{3(P^2 + Q^2)} \Rightarrow \theta_1 = 60^\circ$$

$$\Rightarrow \text{For } |\vec{A}| = |\vec{B}| = \sqrt{2(P^2 + Q^2)}$$

$$\Rightarrow \theta_2 = 90^\circ$$

32. [90°]



So angle between $(2\vec{P} + \vec{Q})$ and $\vec{Q}$ is 90°

Alternate solution

$$|\vec{P} + \vec{Q}| = |\vec{P}|^2 + |\vec{Q}|^2 + 2PQ \cos \theta = P^2$$

$$\Rightarrow Q + 2P \cos \theta = 0 \Rightarrow \cos \theta = -\frac{Q}{2P}$$

$$\frac{2\vec{P} + \vec{Q}}{2\vec{P}} \cdot \frac{\vec{Q}}{2\vec{P}} = 0$$

$$\tan \alpha = \frac{2P \sin \theta}{2P \cos \theta + Q} = \infty$$

$$\alpha = 90^\circ$$

33. (c) $|\vec{A}| = 3, |\vec{A}_1| = 5, |\vec{A} + \vec{A}_1| = 5$

$$|\vec{A} + \vec{A}_1| = \sqrt{|\vec{A}|^2 + |\vec{A}_1|^2 + 2|\vec{A}||\vec{A}_1|\cos\theta}$$

$$5 = \sqrt{9 + 25 + 2 \times 3 \times 5 \cos \theta}$$

$$\Rightarrow \cos \theta = -\frac{3}{10}$$

$$(2\vec{A} + 3\vec{A}_1) \cdot (3\vec{A} - 2\vec{A}_1)$$

$$= 6\vec{A} \cdot \vec{A} + 9\vec{A} \cdot \vec{A}_1 - 4\vec{A} \cdot \vec{A}_1 - 6\vec{A}_1 \cdot \vec{A}_1 = -118.5$$

34. (a) $2|\vec{P} + \vec{Q}| = |\vec{P} + 2\vec{Q}|$

$$\Rightarrow 13 + 12 \cos \theta = 10 + 6 \cos \theta$$

$$\cos \theta = -\frac{1}{2} \Rightarrow \theta = 120^\circ$$

$$35. (4) \vec{B} \times \vec{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 4 & 3 \\ -8 & -1 & 3 \end{vmatrix} = 15\hat{i} - 21\hat{j} + 33\hat{k}$$

$$\vec{A} \cdot (\vec{B} \times \vec{C})$$

$$= (-\hat{i} - 6\hat{j} - 2\hat{k}) \cdot (15\hat{i} - 21\hat{j} + 33\hat{k})$$

$$\text{Since, } \vec{A} \cdot (\vec{B} \times \vec{C}) = 0$$

$$-15x + 126 - 66 = 0$$

$$x = 4$$

$$36. [4] \vec{P} \times \vec{Q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & \sqrt{3} & 2 \\ 4 & \sqrt{5} & 2.5 \end{vmatrix} = \sqrt{3} \frac{\hat{i}}{2} + \frac{\hat{j}}{2} - \sqrt{3} \hat{k}$$

$$\text{and } \Rightarrow |\vec{P} \times \vec{Q}| = 2$$

$$\Rightarrow \frac{\vec{P} \times \vec{Q}}{|\vec{P} \times \vec{Q}|} = \frac{1}{2} \left(\sqrt{3} \frac{\hat{i}}{2} + \frac{\hat{j}}{2} - \sqrt{3} \hat{k} \right)$$

$$= \frac{1}{4} (\sqrt{3}\hat{i} + \hat{j} - 2\sqrt{3}\hat{k}) \Rightarrow x = 4$$

37. (c) Given, $|\vec{A}| \neq 0, \vec{A} \times \vec{A} = |\vec{A}| |\vec{A}| \sin \theta \hat{n}$

$$= |\vec{A}| |\vec{A}| \sin \theta \hat{n} = 0$$

[Since Angle between the vectors are zero degree]

$$\vec{A} \times \vec{A} = 0$$

$$38. (b) \vec{r} = \vec{r} \times \vec{r}$$

$$= (2\hat{i} + \hat{j} + 2\hat{k}) \times (3\hat{i} + 4\hat{j} - 2\hat{k})$$

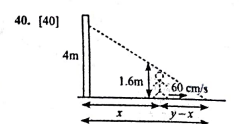
$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 2 \\ 3 & 4 & -2 \end{vmatrix}$$

$$= \hat{i}(-2-8) - \hat{j}(-4-6) + \hat{k}(8-3)$$

$$\vec{r} = -10\hat{i} + 10\hat{j} + 5\hat{k}$$

39. [180] If angle between them is 180° then

$$\vec{P} \times \vec{Q} = \vec{Q} \times \vec{P} = 0$$



From the property of similar triangle,

$$\frac{4}{y} = \frac{1.6}{y-x} \Rightarrow x = 0.6y$$

$$\frac{dx}{dt} = 0.6 \times \frac{dy}{dt} \Rightarrow 60 = 0.6 \times \frac{dy}{dt}$$

$$\therefore \frac{dy}{dt} = 100 \text{ cm/s, i.e., speed of tip of person's shadow w.r.t person} = 100 - 60 = 40 \text{ cm/s}$$

41. (b) $t = \sqrt{x} + 4$, at $t = 4, \sqrt{x} = 0$

$$\text{So, } \frac{dx}{dt} = \frac{1}{2\sqrt{x}} \therefore \left(\frac{dx}{dt} \right)_{t=4} = (2\sqrt{x})_{t=4} = 0$$

42. [3] Direction of $\vec{P}$,

$$\vec{v}_1 = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|} = \frac{(\hat{i} + \hat{j}) \times (\hat{j} + \hat{k})}{|(\hat{i} + \hat{j}) \times (\hat{j} + \hat{k})|} = \frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$$

Direction of $\vec{Q}$,

$$\vec{v}_2 = \frac{\vec{A} \times \vec{C}}{|\vec{A} \times \vec{C}|} = \frac{(\hat{i} + \hat{j}) \times (\hat{j} - \hat{k})}{|(\hat{i} + \hat{j}) \times (\hat{j} - \hat{k})|} = \frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$$

Angle between $\vec{v}_1$ and $\vec{v}_2$,

$$\cos \theta = \frac{\vec{v}_1 \cdot \vec{v}_2}{|\vec{v}_1| |\vec{v}_2|} = \frac{(\hat{i} - \hat{j} + \hat{k}) \cdot (\hat{i} - \hat{j} + \hat{k})}{(\sqrt{3})(\sqrt{3})} = \frac{1}{3}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{3}\right)$$

$$\text{Given, } \theta = \cos^{-1}\left(\frac{1}{\sqrt{x}}\right)$$

$$\therefore x = 3$$